

Programme	B.Sc. Computer Science				
Course Title	Introduction to Data Warehousing and Big Data				
Type of Course	Elective				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-		60
Pre-requisites	1. Data Science Concepts 2. RDBMS				
Course Summary	This course provides insight into the basic concepts of data warehousing and its architecture. The various OLAP operations are also discussed in this syllabus to understand the summarisation and retrieval of the data. The fundamentals of big data technology are also introduced in this syllabus following the data warehousing concepts. An overview of the storage, retrieval and processing of big data is also provided here.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts of data warehouse and its architecture	U	C	Instructor-created exams / Quiz
CO2	Analyse the differences between OLTP and OLAP operations	An	C	Instructor-created exams / Quiz
CO3	Understand the various operations performed in the data warehouse to process the data	U	C	Modelling Assignments/ Case Studies
CO4	Understand Big Data and the importance of cloud and distributed computing in the real world	U	C	Instructor-created exams / Quiz
CO5	Understand the Map Reduce concepts of the jobs	U	C	Modelling Assignments/ / Case studies
CO5	Understand the Hadoop ecosystem	U	C	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Data Warehousing		8	10
	1	Overview of databases and need for normalisation. Databases vs data warehouse	2	
	2	Introduction to Data warehousing, Need for data warehousing	2	
	3	Architecture of data warehousing	3	
	4	Data Marts vs Data Lakes	1	
II	Concepts and techniques in Data Warehousing		13	20
	6	Data warehouse Schema - Stars, snowflakes and fact constellations	3	
	7	OLAP (Online analytical processing) definitions// Difference between OLAP and OLTP	2	
	8	Dimensional analysis - What are cubes?	2	
	9	Drill-down and roll-up - slice and dice or rotation	2	
	10	OLAP models, ROLAP versus MOLAP	4	
III	Big Data Technology		16	25
	11	Fundamentals of Big Data, 3V's of big data. Structured Data and its sources; Unstructured data and its sources; integrating data types to big data	2	
	12	Big Data Stack: Layers 1 to 4; Big data analytics and applications	4	
	13	Role of Distributed computing and virtualizations in big data	3	
	14	Hypervisor and implementing virtualizations in big data	1	
	15	Cloud in big data; cloud deployment models	2	
	16	Cloud delivery models; advantages of using cloud	2	
	17	Cloud Providers for Big Data	2	
IV	Big Data Management		11	15
	18	Fundamentals of Map Reduce: Map and reduce functions	2	
	19	Putting Map and Reduce together	2	
	20	Hadoop: Name nodes, Data Nodes, Hadoop MapReduce	3	
	21	Hadoop ecosystem: Yarn, HBase and Hive Interactive tools: Pig, Pig Latin, SQOOP, ZooKeeper	4	
	22	Big Data Analytics: Basic, Advanced, Operationalized	1	
V	Open ended module - Hands-on data and Data warehousing: Practical Applications, Case Study and Course Project		12	
	1	Data warehousing case studies	4	
	2	Case studies on Big Data Analytics and Big Data Solutions in the Real World	5	
	3	Assignments on Security in Big Data Environments	3	

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	-	-	-	-	1						
CO 2	1	-	2	-	-	-						
CO 3	-	-	2	-	-	-						
CO 4	-	2	3	3	-	1						
CO 5	-	2	3	3	-	1						
CO 6	-	-	-	-	-	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1		✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓			✓
CO 5	✓		✓	✓
CO 6	✓		✓	✓

References

1. O'Neil, Cathy, and Rachel Schutt. *Doing data science: Straight talk from the frontline*. " O'Reilly Media, Inc.", 2013.
2. Han, Jiawei, et al. *Data Mining: Concepts and Techniques*. Netherlands, Elsevier Science, 2011.
3. Shah, Chirag. *A Hands-On Introduction to Data Science*. United Kingdom, Cambridge University Press, 2020.
4. Chopra, Rohan, et al. *Data Science with Python: Combine Python with Machine Learning Principles to Discover Hidden Patterns in Raw Data*. United Kingdom, Packt Publishing, 2019.